3-31-23 — 4-3-23

Friday, March 31, 2023 10:26 AM

what: $A = QR$ Q: unitary. orthogonal.
R: upper triangular

why: least squares in a numerically stable way.

How: G-S, MGS. ← we know this
Householder ← new
Given rotations ← new } Numerically Stable

I) General strategy

A) A little observation

Q_1 : unitary $Q_1^H Q_1 = I$

Q_2 : unitary $Q_2^H Q_2 = I$

$$Q = Q_1 Q_2$$

$$Q^H Q = (Q_1 Q_2)^H (Q_1 Q_2)$$
$$= \cancel{Q_2^H} \cancel{Q_1^H} \cancel{Q_1} \cancel{Q_2}$$
$$= I$$

B) Strategy

$$\underbrace{(Q_n \cdots Q_2 Q_1)}A = R = \begin{bmatrix} \ddots \end{bmatrix}$$

The question is how to find this

$$\underbrace{(Q_1^H Q_2^H \cdots Q_{n-1}^H Q_n^H)}_{Q} R$$

Applying the Q matrices will make A upper triangular.

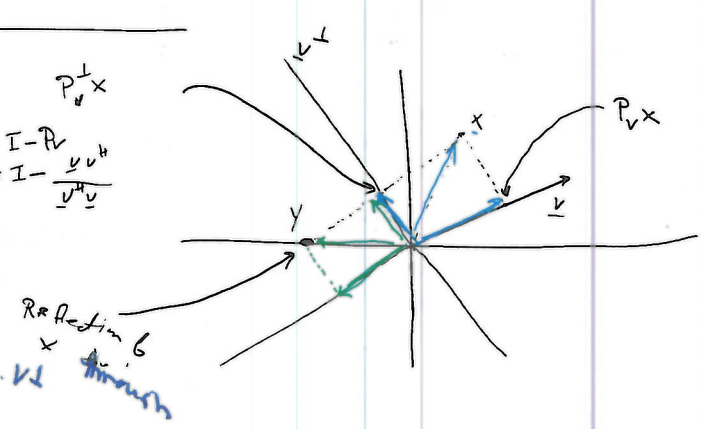
$$\underline{A = QR}$$

II) Householder

The big idea is that we will zero out columns to incrementally generate an upper triangular matrix.

$P_v^\perp x$

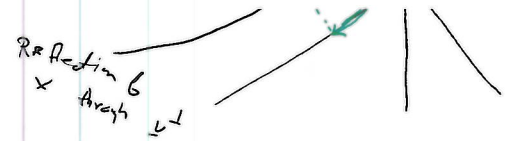
$P_v^\perp = I - P_v$
$= I - \frac{v v^H}{v^H v}$



$P_v = \frac{v v^H}{v^H v}$

This is dependent on v, so it is important to get the right v.

$x = P_v x + P_v^\perp x$
$= P x + (I - P_v) x$



$$\underline{x} = P_v x + P_v^\perp x$$
$$= P_v x + (I - P_v)x$$

$$y = P_v^\perp x + (-P_v \underline{x})$$
$$= \left[\left(I - \frac{\underline{v}\,\underline{v}^H}{\underline{v}^H\underline{v}}\right) - \frac{\underline{v}\,\underline{v}^H}{\underline{v}^H\underline{v}}\right]\underline{x}$$
$$= \left(I - 2\frac{\underline{v}\,\underline{v}^H}{\underline{v}^H\underline{v}}\right)\underline{x}$$

H_v is unitary

Note: $H_v^H = H_v$

$y = H_v \underline{x}$ — Our reflection

1) Is this a reflection?

- If we "reflect" this again, do we get x back?

$$H_v H_v \underline{x} = \left(I - \frac{2\underline{v}\,\underline{v}^H}{\underline{v}^H\underline{v}}\right)\left(I - \frac{2\underline{v}\,\underline{v}^H}{\underline{v}^H\underline{v}}\right)x$$
$$= I - \frac{2\underline{v}\,\underline{v}^H}{\underline{v}^H\underline{v}} - \frac{2\underline{v}\,\underline{v}^H}{\underline{v}^H\underline{v}} + 4\frac{\underline{v}(\cancel{\underline{v}^H\underline{v}})\underline{v}^H}{\underline{v}^H\underline{v}\,\cancel{\underline{v}^H\underline{v}}}$$
$$= I - \frac{4\underline{v}\,\underline{v}^H}{\underline{v}^H\underline{v}} + \frac{4\underline{v}\,\underline{v}^H}{\underline{v}^H\underline{v}}$$
$$= I$$

yes

2) Another check

$$\|H_v x\|^2 = \langle H_v x, H_v x\rangle = x^H H_v^H H_v x$$
$$= x^H H_v H_v x$$
$$= \underline{x}^H \underline{x}$$

What this does is it takes the energy of x and puts it into the first component

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A) How to get our V.

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

All the energy goes into the first coefficient.

$$H_v x = \begin{bmatrix} \alpha \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\|x\|_2^2 = \|H_v x\|_2^2 = |\alpha|^2$$
$$\alpha = \pm\|x\|_2$$

$$H_v x = \alpha \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \alpha \underline{e}_1$$

Write out H_v as we did before

Find a $\underline{v}$ such that

$$\left(I - \frac{2\underline{v}\,\underline{v}^H}{\underline{v}^H\underline{v}}\right)\underline{x} = \alpha \underline{e}_1$$

We don't care about magnitude, just direction.

$$\underline{x} - \frac{2\underline{v}\overbrace{(\underline{v}^H\underline{x})}^{\text{scalar}}}{\underline{v}^H\underline{v}} = \alpha \underline{e}_1$$

Because this is a scalar, it can be moved around.

Vector V is proportional to $\left(\frac{2v^H x}{v^H v}\right)$ so we care about

$$\left(2\frac{v^H x}{v^H v}\right) v = x - \alpha e_1$$

Take $v = x - \alpha e_1$

$$\boxed{v = x \pm \|x\|_2 e_1}$$

$$\boxed{v = x + \text{sign}(x(1))\|x\|_2 e_1}$$

If $x \approx \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$ is close to e_1

This is bad to subtract

So we want to add.

If $x \approx -e_1$, then we want to subtract.

To fix numerical issues, we write this

Because of this

All this was for the first column

Expand

B) Build the matrix

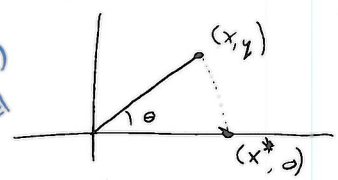
$$\begin{bmatrix} 1 & 0 \\ 0 & H_2 \end{bmatrix} H_1 A = \begin{bmatrix} \alpha_1 & x & x \\ 0 & x & x \\ 0 & x & x \\ \vdots & \vdots & \\ 0 & x & x \end{bmatrix}$$

H_1

$$\begin{bmatrix} 1 & & 0 \\ & 1 & \\ 0 & & H_3 \end{bmatrix} \qquad \begin{bmatrix} \alpha_1 & x & x & x \\ 0 & \alpha_2 & x & x \\ 0 & 0 & \cdot & \\ \vdots & & \cdot & \\ 0 & 0 & x & x \end{bmatrix}$$

III) Givens Rotations

Like Householder, we are trying to get a vector to be placed on an axis. This time, we are doing it with rotations instead of ~~reflec~~ projected.



Rotation: $x^2 + y^2 = x^{*2}$

The rotation does not change the energy

$$G_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

rotates an angle θ

$\theta > 0$: ccw

$\theta < 0$: cw

$$\theta = -\tan^{-1}\frac{y}{x}$$

$$\underbrace{\begin{bmatrix} 1 & & \\ & 1 & \\ & & \end{bmatrix}}_{Q} \overbrace{\begin{bmatrix} x & x & x \\ x & x & x \end{bmatrix}}^{A}$$

Similarly to Householder, we will apply a rotation matrix to incrementally create our upper diagonal matrix.

Angles that we compute

$$\theta = -\tan^{-1}\left(\frac{y}{x}\right)$$

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & & G_\theta \end{bmatrix} \begin{bmatrix} x & x & x \\ x & x & x \\ \begin{bmatrix} x^x \\ x^y \end{bmatrix} & x & x \\ & x & x \end{bmatrix}$$

Q_2

$$\begin{bmatrix} 1 & & \\ & G_\theta & \\ & & 1 \end{bmatrix} \begin{bmatrix} x & x & x \\ \begin{bmatrix} x \\ x^* \end{bmatrix}^{x,y} & x & x \\ 0 & x & x \end{bmatrix}$$

Q_3

$$\begin{bmatrix} G_\theta & & \\ & 1 & \\ & & 1 \end{bmatrix} \begin{bmatrix} \begin{bmatrix} x \\ x^* \end{bmatrix}^{x,y} & x & x \\ 0 & x & x \\ 0 & x & x \end{bmatrix}$$

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & & G_\theta \end{bmatrix} \begin{bmatrix} x^* & x & x \\ 0 & x & x \\ 0 & \begin{bmatrix} x \\ x \end{bmatrix}^{x,y} & x \\ 0 & & x \end{bmatrix} \qquad (Q_3 Q_2 Q_1)$$

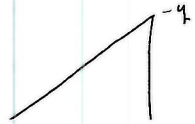
$$\begin{bmatrix} x^* & x & x \\ 0 & \begin{bmatrix} x \\ x^* \end{bmatrix} & x \\ 0 & & x \\ 0 & 0 & x \end{bmatrix} \qquad Q_5 Q_4$$

Q_6

$$\underbrace{(Q_6 Q_5 Q_4 Q_3 Q_2 Q_1)}_{Q^H} \begin{bmatrix} & A & \end{bmatrix} = \overbrace{\begin{bmatrix} x & x & x \\ 0 & x & x \\ 0 & 0 & x \\ 0 & 0 & 0 \end{bmatrix}}^{R}$$

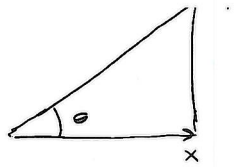
Unitary

$$\rightarrow \boxed{A = QR}$$



$$\theta = \tan^{-1}\left(\frac{-y}{\ }\right)$$

A) Trig Facts to Simplify



$$\theta = \tan^{-1}\left(\frac{-y}{x}\right)$$

$$\cos\theta = \cos\tan^{-1}\left(\frac{-y}{x}\right) = \frac{x}{\sqrt{x^2+y^2}}$$

$$\sin\theta = \sin\tan^{-1}\left(\frac{-y}{x}\right) = \frac{-y}{\sqrt{x^2+y^2}}$$

$$\tan\theta = \frac{\sin\theta}{\cos\theta}$$

$$\theta = \tan^{-1}\left(\frac{\sin\theta}{\cos\theta}\right)$$

Plug in

Chapter 6

I) Eigen- characteristic

$$y_1[t+1] = -y_1[t] - 1.5\,y_2[t]$$
$$y_2[t+1] = 0.5\,y_1[t] + y_2[t]$$

Try: $\begin{bmatrix} y_1[t] = \lambda^t x_1 \\ y_2[t] = \lambda^t x_2 \end{bmatrix}$ $\begin{bmatrix} y_1[t] \\ y_2[t] \end{bmatrix} = \lambda^t \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

Plug in to get the system

$$\lambda^{t+1}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -\lambda^t x_1 - 1.5\lambda^t x_2 \\ 0.5\lambda^t x_1 + 1\lambda^t x_2 \end{bmatrix}$$

Simplify

$$\lambda\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \underbrace{\begin{bmatrix} -1 & -1.5 \\ .5 & 1 \end{bmatrix}}_{A}\underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\underline{x}}$$

Now we have this equation which looks awfully familiar.

$$\boxed{A\underline{x} = \lambda\underline{x}}$$

eigen equation

$\underline{x}$: eigenvector

λ: eigenvalue.

warm-up

$$y[t+1] = -.5\,y[t]$$
$$y[0] = 1$$
$$y[t] = c(-0.5)^t$$

Try: $y[t] = \lambda^t$, $y[t+1] = \lambda^{t+1}$

$$\lambda^{t+1} = (-.5)\lambda^t$$
$$\lambda = -.5$$
$$\underline{\lambda + .5 = 0}$$

$$(A - \lambda I)\underline{x} = 0$$
$$(\lambda I - A)\underline{x} = 0$$
$$\det(A - \lambda I) = 0$$
$$\underbrace{\underbrace{\det(\lambda I - A)}_{\text{characteristic polynomial of } A} = 0}_{\text{characteristic equation of } A}$$

A) Get the eigen stuff

$$A = \begin{bmatrix} -1 & -1.5 \\ .5 & 1 \end{bmatrix}$$

$$(\lambda I - A) = \begin{bmatrix} \lambda+1 & 1.5 \\ -.5 & \lambda-1 \end{bmatrix}$$

1) Eigenvalues

$$\det(\lambda I - A) = (\lambda+1)(\lambda-1) + \frac{3}{4}$$

$$= \lambda^2 - 1 + \frac{3}{4}$$

$$= \lambda^2 - .25 \qquad \lambda^2 - .25 = 0$$

$$\left.\begin{array}{l}\lambda = 1/2 \\ \lambda = -1/2\end{array}\right\} \text{Eigenvalues}$$

2) Eigen vectors

$\underline{\underline{\lambda = \frac{1}{2}}}$ $\quad \lambda I - A = \begin{bmatrix} 1/2 & 3/2 \\ -1/2 & -1/2 \end{bmatrix}$ $\qquad (\lambda I - A)\underline{x} = \underline{0}$

$\underline{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \underline{\underline{x_1}}$ $\qquad \underline{x} = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$ $\quad \begin{bmatrix} 65 \\ -65 \end{bmatrix}$

$\underline{\underline{\lambda = -1/2}}$ $\quad \lambda I - A = \begin{bmatrix} 1/2 & \frac{3}{2} \\ -\frac{1}{2} & -\frac{3}{2} \end{bmatrix}$ $\qquad \underline{x} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}$ $\quad \underline{x} = \begin{bmatrix} -3 \\ 1 \end{bmatrix}$ $\quad \underline{x} = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$

$\underline{\underline{x_2}}$

B) Back to the D.E.:

Solution will be of the form

$$\begin{bmatrix} y_1[t] \\ y_2[t] \end{bmatrix} = \lambda^t \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Take a linear combination of our eigen stuff

$$\boxed{\begin{bmatrix} y_1[t] \\ y_2[t] \end{bmatrix} = c_1 \underline{\left(\frac{1}{2}\right)}^t \underline{x_1} + c_2 \underline{\left(\frac{-1}{2}\right)}^t \underline{\underline{x_2}}}$$

Recall:

$$x[t+1] = A\,x[t] + B\,u[t]$$

$$y[t] = C\,x[t]$$

$$H(z) = C(zI - A)^{-1} B$$

$$= C \frac{\operatorname{adj}(zI - A)}{\det(zI - A)} B$$

poles of T.F. = eigenvalues of A.